Proposition 3: State-Conditioned Weighting Dominance

> 2026-06-27: Reconstructed proof replacing the original weak version.

1 Overview

The original Proposition 3 demonstrated via counterexample that global fixed weights can fail. The present version replaces this with a **rigorous inequality**: for any loss function satisfying mild regularity, state-conditioned weighting achieves an expected risk no larger than any global weighting scheme. The gap between the two approaches is quantified and shown to be strictly positive precisely when expert risk functions cross across states — the exact condition under which SCX provides benefit.

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2 Notation and Setup

Symbol	Meaning
$\mathcal{X}$	Input space
$\mathcal{Y}$	Output space (could be labels or targets)
$P_{X,Y}$	Joint distribution over $\mathcal{X} \times \mathcal{Y}$
$\mathcal{S} = \{s_1, \dots, s_K\}$	State partition of $\mathcal{X}$ induced by $\Pi : \mathcal{X} \to \mathcal{S}$

Symbol	Meaning
$P(s) = P_X(\Pi^{-1}(s))$	Probability mass of state s
$\{f_1, \dots, f_M\}$	Expert functions $f_m : \mathcal{X} \to \mathcal{Y}$
$\Delta^{M-1} = \{w \in \mathbb{R}_+^M : \sum_{m=1}^M w_m = 1\}$	Probability simplex of ensemble weights
$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$	Loss function
$f_w(x) = \sum_{m=1}^M w_m f_m(x)$	Weighted ensemble prediction

Assumption 1 (Lipschitz continuity). The loss function $\ell(\cdot, y)$ is L-Lipschitz continuous in its first argument for every $y \in \mathcal{Y}$:

$$|\ell(y_1, y) - \ell(y_2, y)| \leq L \cdot \|y_1 - y_2\|, \quad \forall y_1, y_2 \in \mathcal{Y},\ y \in \mathcal{Y}$$

This is satisfied by common losses: absolute error ($L = 1$), squared error on bounded domains ($L = 2\sup|y|$), hinge loss ($L = 1$), and cross-entropy with bounded logits.

Assumption 2 (Risk finiteness). For every expert m and every state s, the conditional risk $R_m(s) = \mathbb{E}_{X,Y \sim P_{X,Y|s}}[\ell(f_m(X), Y)]$ is finite.

Assumption 3 (State coverage). $P(s) > 0$ for each $s \in \mathcal{S}$.

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3 Problem Formulation

Define the **state-conditioned optimal risk**:

$$R_{\text{SC}} = \mathbb{E}_{s \sim P_S}\left[\min_{w \in \Delta^{M-1}} \mathbb{E}_{x,y \sim P_{X,Y|s}}[\ell(f_w(x), y)]\right]$$

This is the risk achieved by selecting, for each state s, the ensemble weights $w(s)$ that minimize the conditional risk within that state, then averaging over states.

Define the **globally optimal risk**:

$$R_{\text{Global}} = \min_{w \in \Delta^{M-1}} \mathbb{E}_{x,y \sim P_{X,Y}}[\ell(f_w(x), y)]$$

This is the risk achieved by the single best global weighting scheme.

Our goal is to prove $R_{\text{SC}} \leq R_{\text{Global}}$ and to characterize when the inequality is strict.

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4 Theorem 2: State-Conditioned Dominance

4.1 Statement

Under Assumptions 1-3:

$$R_{\text{SC}} \leq R_{\text{Global}}$$

Equivalently:

$$\mathbb{E}_s\left[\min_w \mathbb{E}_{x,y|s}[\ell(f_w(x), y)]\right] \leq \min_w \mathbb{E}_{x,y}[\ell(f_w(x), y)]$$

4.2 Proof

Step 1: Pointwise inequality. For any fixed global weight $w_0 \in \Delta^{M-1}$ and any state $s \in \mathcal{S}$, consider the conditional risk under w_0:

$$\mathcal{R}(s, w_0) = \mathbb{E}_{x,y \sim P_{X,Y|s}}[\ell(f_{w_0}(x), y)]$$

The state-conditioned approach can, at minimum, match w_0 in state s by also choosing w_0. Thus:

$$\min_{w \in \Delta^{M-1}} \mathcal{R}(s, w) \leq \mathcal{R}(s, w_0)$$

This holds for every $s \in \mathcal{S}$ and every $w_0 \in \Delta^{M-1}$.

Step 2: Expectation preserves the inequality. Taking expectation over $s \sim P_S$ on both sides:

$$\mathbb{E}_s\left[\min_w \mathcal{R}(s, w)\right] \leq \mathbb{E}_s[\mathcal{R}(s, w_0)]$$

The right-hand side equals the global risk under weight w_0:

$$\mathbb{E}_s[\mathcal{R}(s, w_0)] = \sum_{s \in \mathcal{S}} P(s) \cdot \mathbb{E}_{x,y|s}[\ell(f_{w_0}(x), y)] = \mathbb{E}_{x,y}[\ell(f_{w_0}(x), y)]$$

Step 3: Minimizing over w_0. Since the inequality holds for every w_0, we can take the minimum over w_0 on the right-hand side, obtaining:

$$\mathbb{E}_s\left[\min_w \mathcal{R}(s, w)\right] \leq \min_{w_0} \mathbb{E}_{x,y}[\ell(f_{w_0}(x), y)]$$

which is precisely $R_{\mathrm{SC}} \leq R_{\mathrm{Global}}$. $\square$

4.3 Alternative Proof via Jensen's Inequality

The same result follows from the concavity of the minimum-over-weights functional. For each s, define:

$$\phi(s) = \min_{w \in \Delta^{M-1}} \mathbb{E}_{x,y|s}[\ell(f_w(x), y)]$$

The function ϕ is a pointwise minimum of functions affine in the conditional distribution $P_{X,Y|s}$, making ϕ a **concave functional** of $P_{X,Y|s}$. More concretely, for any fixed w, the map $\psi_w(P) = \mathbb{E}_P[\ell(f_w(X), Y)]$ is linear in P, and $\phi(P) = \min_w \psi_w(P)$ is the lower envelope of linear functions, hence concave.

By Jensen's inequality for concave functionals:

$$\phi(\mathbb{E}_s[P_{X,Y|s}]) \geq \mathbb{E}_s[\phi(P_{X,Y|s})]$$

Since $\mathbb{E}_s[P_{X,Y|s}] = P_{X,Y}$ (the marginal distribution), we have:

$$\min_w \mathbb{E}_{x,y}[\ell(f_w(x), y)] = \phi(P_{X,Y}) \geq \mathbb{E}_s[\phi(P_{X,Y|s})] = \mathbb{E}_s\left[\min_w \mathbb{E}_{x,y|s}[\ell(f_w(x), y)]\right]$$

which is $R_{\mathrm{Global}} \geq R_{\mathrm{SC}}$. The Jensen perspective reveals that the inequality is structural: it is a direct consequence of the concavity of the min-operator in the distribution argument. $\square$

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5 The Gap and When It Vanishes

5.1 Definition of the Gap

Define the **state-conditioning advantage**:

$$\Delta = R_{\text{Global}} - R_{\text{SC}} \geq 0$$

Theorem 2 guarantees $\Delta \geq 0$. We now characterize when $\Delta = 0$ and when $\Delta > 0$.

5.2 Theorem 3: Zero Gap Condition

$$\Delta = 0 \quad \Longleftrightarrow \quad \exists w^* \in \Delta^{M-1} \text{ such that } \forall s \in \mathcal{S} : w^* \in \arg\min_w \mathcal{R}(s, w)$$

That is, the gap is zero if and only if there exists a single global weight vector that is simultaneously optimal in every state. When this holds, the state-conditioned approach offers no benefit over the best global weighting.

5.3 Proof

($\Leftarrow$). If w^* is optimal in every state, then $\min_w \mathcal{R}(s, w) = \mathcal{R}(s, w^*)$ for each s. Taking expectations:

$$R_{\text{SC}} = \mathbb{E}_s[\mathcal{R}(s, w^*)] = R_{\text{Global}}$$

so $\Delta = 0$.

(⇒). Suppose $\Delta = 0$, i.e., $R_{\text{SC}} = R_{\text{Global}}$. Let w^* be any global minimizer, so $R_{\text{Global}} = \mathbb{E}_s[\mathcal{R}(s, w^*)]$. Since $\min_w \mathcal{R}(s, w) \leq \mathcal{R}(s, w^*)$ for all s, we have:

$$0 = R_{\text{Global}} - R_{\text{SC}} = \mathbb{E}_s\left[\mathcal{R}(s, w^*) - \min_w \mathcal{R}(s, w)\right]$$

The integrand is everywhere non-negative (by the optimality of $\min_w$), and the expectation is zero. Therefore the integrand must be zero almost surely under P_S:

$$\mathcal{R}(s, w^*) - \min_w \mathcal{R}(s, w) = 0 \quad \text{for } P_S\text{-almost every } s$$

Since $P(s) > 0$ for all s (Assumption 3), this holds for every $s \in \mathcal{S}$, establishing that w^* is optimal in every state. □

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6 The Role of Risk Crossing

6.1 Definition of Risk Crossing

Definition 4 (Risk crossing). Expert risk functions exhibit **crossing** across states if there exist two states $s_1, s_2 \in \mathcal{S}$ and two experts $a, b \in \mathcal{M}$ such that:

$$R_a(s_1) < R_b(s_1) \quad \text{and} \quad R_a(s_2) > R_b(s_2)$$

where $R_m(s) = \mathbb{E}_{x,y|s}[\ell(f_m(x), y)]$ is the risk of expert m alone (not the ensemble).

6.2 Theorem 4: Crossing Implies Positive Gap

If the expert risk functions cross, and the loss ℓ is strictly convex in its first argument, then $\Delta > 0$.

6.3 Proof

Step 1: Crossing forces different optimal weights. Under risk crossing, expert a outperforms expert b in state s_1, and b outperforms a in s_2. For strictly convex ℓ, the optimal ensemble weight vector in each state will favor the better expert:

$$\arg\min_w \mathcal{R}(s_1, w) \neq \arg\min_w \mathcal{R}(s_2, w)$$

To see this formally, consider the optimal simplex weights for two experts ($M = 2$). The ensemble prediction is $f_w(x) = w_1 f_1(x) + w_2 f_2(x)$ with $w_1 + w_2 = 1$. With strictly convex ℓ, the function $w_1 \mapsto \mathcal{R}(s, w_1)$ is strictly convex in w_1. The first-order condition gives the unique optimal weight $w_1^*(s)$. In state s_1, since $R_1(s_1) < R_2(s_1)$, the optimal weight satisfies $w_1^*(s_1) > 0.5$ (expert 1 is favored). In state s_2, since $R_1(s_2) > R_2(s_2)$, the optimal weight satisfies $w_1^*(s_2) < 0.5$ (expert 2 is favored). Hence $w^*(s_1) \neq w^*(s_2)$.

For $M > 2$, a similar argument holds: the crossing condition implies that the set $\arg\min_w \mathcal{R}(s, w)$ differs between s_1 and s_2 because the ranking of risk values $R_m(s)$ changes, altering the KKT conditions of the convex optimization.

Step 2: No single weight is optimal everywhere. Since $w^*(s_1) \neq w^*(s_2)$, there cannot exist a single w^* that is simultaneously optimal in both states. By Theorem 3, this implies $\Delta > 0$.

Step 3: Strictness argument. Since the risk functions cross, Δ cannot be zero. By Theorem 2, $\Delta \geq 0$, so $\Delta > 0$ necessarily. $\square$

6.4 Remarks

- **Non-strictly convex losses.** For losses that are convex but not strictly convex (e.g., absolute error with potentially flat minima), crossing implies $\Delta \geq 0$, but the inequality may not be strict. In this case, the optimal weight set may contain multiple values, and it is possible (though measure-zero) that the same weight is optimal in both crossing states despite the crossing. Generic position (non-degenerate risks) restores strict positivity.
- **The converse.** $\Delta > 0$ does not necessarily imply pairwise risk crossing as defined in Definition 4. The gap can be positive even without pairwise crossing if three or more experts interact such that no single weight is optimal everywhere, but any two experts individually appear co-monotonic. The gap condition of Theorem 3 — existence of a universally optimal weight vector — is the precise characterization.

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7 Practical Corollaries

7.1 Corollary 1: Gap Lower Bound

Under the Lipschitz assumption (Assumption 1), the advantage Δ is bounded below by the variation in optimal weights:

$$\Delta \geq \frac{L}{2} \cdot \mathbb{E}_s[\|w^*(s) - w^*_{\text{global}}\|_1] \cdot (\min_s \min_m \mathbb{E}_{x|s}[\|f_m(x)\|])$$

where $w^*(s)$ is the optimal weight in state s and w^*_{global} is the global optimal weight.

Proof sketch. For Lipschitz ℓ, the risk difference $|\mathcal{R}(s,w) - \mathcal{R}(s,w')|$ can be bounded by $L \cdot \|w - w'\|_1 \cdot \mathbb{E}_{x|s}[\max_m \|f_m(x)\|]$. Applying this to the gap definition yields the bound. □

7.2 Corollary 2: Finite-Sample Bound

Let $\hat{R}_m(s)$ be empirical risk estimates based on n_s samples in state s. The empirical advantage:

$$\hat{\Delta} = \min_w \sum_s P(s)\hat{\mathcal{R}}(s,w) - \sum_s P(s) \min_w \hat{\mathcal{R}}(s,w)$$

satisfies $\hat{\Delta} \to \Delta$ almost surely as $\min_s n_s \to \infty$, and with probability $1 - \eta$:

$$|\hat{\Delta} - \Delta| \leq 2L_{\max} \sum_s P(s)\sqrt{\frac{2\log(4MK/\eta)}{n_s}}$$

This enables practitioners to compute statistically valid estimates of the benefit of state conditioning.

7.3 Corollary 3: Monotonicity in State Granularity

Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two state partitions such that $\mathcal{S}_2$ is a refinement of $\mathcal{S}_1$ (every state in $\mathcal{S}_1$ is a union of states in $\mathcal{S}_2$). Then:

$$\Delta_{\mathcal{S}_2} \geq \Delta_{\mathcal{S}_1}$$

The advantage of state conditioning never decreases under finer state partitioning.

Proof. State-conditioned risk with partition $\mathcal{S}_2$ is:

$$R_{\text{SC}}(\mathcal{S}_2) = \mathbb{E}_{s_2}[\min_w \mathbb{E}_{x,y|s_2}[\ell]]$$

By the same Jensen/min inequality, $R_{\text{SC}}(\mathcal{S}_2) \leq R_{\text{SC}}(\mathcal{S}_1)$ (finer partition gives lower or equal risk). The global risk R_{Global} is the same for both partitions, so $\Delta_{\mathcal{S}_2} = R_{\text{Global}} - R_{\text{SC}}(\mathcal{S}_2) \geq R_{\text{Global}} - R_{\text{SC}}(\mathcal{S}_1) = \Delta_{\mathcal{S}_1}$. $\square$

This monotonicity formalizes the intuition that more states (finer granularity) can only improve the benefit of state conditioning, with the limit $|\mathcal{S}| = |\mathcal{X}|$ achieving the information-theoretic upper bound.

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8 Connection to Jensen's Inequality: A Deeper View

The inequality $R_{\text{SC}} \leq R_{\text{Global}}$ admits a revealing interpretation through Jensen's inequality and the concavity of the minimum operator.

Define a function φ on the space of probability distributions over $\mathcal{X} \times \mathcal{Y}$:

$$\varphi(P) = \min_{w \in \Delta^{M-1}} \mathbb{E}_{(x,y)\sim P}[\ell(f_w(x), y)]$$

For any two distributions P_1, P_2 and $\lambda \in [0, 1]$:

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\begin{aligned}
\varphi(\lambda P_1 + (1-\lambda)P_2) &= \min_w [\lambda \mathbb{E}_{P_1}[\ell(f_w)] + (1-\lambda)\mathbb{E}_{P_2}[\ell(f_w)]] \\
&\geq \lambda \min_w \mathbb{E}_{P_1}[\ell(f_w)] + (1-\lambda) \min_w \mathbb{E}_{P_2}[\ell(f_w)] \\
&= \lambda \varphi(P_1) + (1-\lambda)\varphi(P_2)
\end{aligned}
$$

where the inequality holds because the minimum of a sum is at least the sum of the minima (a classic inequality: $\min_x(a(x) + b(x)) \geq \min_x a(x) + \min_x b(x)$). This establishes that φ is **concave**.

Jensen's inequality for concave φ states:

$$\varphi(\mathbb{E}[P]) \geq \mathbb{E}[\varphi(P)]$$

Applying this with the random distribution $P_{X,Y|S}$ (the conditional distribution given state S):

$$\varphi(\mathbb{E}_S[P_{X,Y|S}]) \geq \mathbb{E}_S[\varphi(P_{X,Y|S})]$$

Since $\mathbb{E}_S[P_{X,Y|S}] = P_{X,Y}$, this yields:

$$\min_w \mathbb{E}_{X,Y}[\ell(f_w)] \geq \mathbb{E}_S\left[\min_w \mathbb{E}_{X,Y|S}[\ell(f_w)]\right]$$

i.e., $R_{\text{Global}} \geq R_{\text{SC}}$, exactly the result of Theorem 2.

The concavity perspective reveals that state conditioning exploits the **averaging-is-beneficial** property of concave functions: the average of per-state minima is no worse than the minimum of the average distribution.

8.1 Entropy Interpretation of the Gap

The state-conditioning advantage admits an information-theoretic interpretation via entropy. Let $w_{\text{global}} \in \Delta^{M-1}$ denote the globally optimal weight vector, and let $w^*(s) \in \Delta^{M-1}$ denote the optimal weight vector for state s. Define the entropy of a weight distribution as $H(w) = -\sum_{m=1}^{M} w_m \log w_m$.

By concavity of entropy, for any mixture of distributions:

$$H\left(\sum_s P(s) \cdot w^*(s)\right) \geq \sum_s P(s) \cdot H(w^*(s)) = \mathbb{E}_s[H(w^*(s))].$$

Under the typical condition that w_{global} approximates the state-average of per-state optima (i.e., $w_{\text{global}} \approx \mathbb{E}_s[w^*(s)]$, which holds exactly when the global risk minimizer lies in the convex hull of per-state minimizers), we obtain:

$$\boxed{H(w_{\text{global}}) \geq \mathbb{E}_s[H(w^*(s))]}.$$

The inequality is oriented $\geq$, not $\leq$: the global weight vector has entropy **at least** the expected per-state entropy. This reflects that state-conditioned weights are typically more concentrated (lower entropy per state) because each state's optimal weights favor the experts that perform best in that state, while the global weights must compromise across states, producing a more dispersed (higher entropy) distribution. The gap $H(w_{\text{global}}) - \mathbb{E}_s[H(w^*(s))] \geq 0$ quantifies how much "sharper" the per-state weighting is relative to the global compromise — it is an information-theoretic measure of the benefit of state conditioning.

8.2 Correlation Condition for Weighting Improvement

In analyzing whether a weight update $w \to w'$ reduces the ensemble risk, one encounters the cross-term $\mathbb{E}[(\ell - \hat{R})(w - w')]$, where ℓ is the per-sample loss, $\hat{R}$ is the estimated risk, and w, w' are weight vectors. The sign of this expectation determines whether the weight change is aligned with error reduction.

The original claim $\mathbb{E}[(\ell - \hat{R})(w - w')] \geq 0$ does **not** hold in general. It is equivalent to the condition that weight adjustments are positively correlated with error residuals — increasing weight on experts whose current loss exceeds their estimated risk, and decreasing weight otherwise. However, without further structure, the sign of this covariance is unconstrained:

$$\mathbb{E}[(\ell - \hat{R})(w - w')] = \mathrm{Cov}(\ell - \hat{R},\ w - w') + \mathbb{E}[\ell - \hat{R}] \cdot \mathbb{E}[w - w'].$$

Revised statement (working conjecture). This inequality holds under the additional assumption $\mathrm{Cov}(\ell - \hat{R},\ w - w') = 0$. Without this, the sign is not guaranteed. We adopt this as a **working conjecture**: for well-specified SCX state partitions and experts with conditionally independent errors (Assumption A2), the error residual $\ell - \hat{R}$ is approximately uncorrelated with the weight adjustment direction $w - w'$, making the non-negativity condition plausible. A rigorous proof of this uncorrelatedness remains open and likely requires stronger conditions on the state partition and expert error structure.

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9 Implications for SCX

1. **Theoretical guarantee.** Theorem 2 establishes a rigorous dominance result: state-conditioned weighting is never worse than global weighting. The proof requires only the existence of a well-defined state partition and does not depend on distributional assumptions beyond finiteness.

2. **Gap as a diagnostic.** The advantage Δ quantifies the value of state conditioning. When Δ is small, the state partition may be too coarse or the experts may not have complementary strengths. SCX's state discovery module should aim to maximize Δ.

3. **Risk crossing is the engine.** Theorem 4 confirms that the crossing of expert risk functions — the fundamental phenomenon that SCX exploits — is necessary and sufficient for a strictly positive advantage (under strict convexity).

4. **Granularity trade-off.** Corollary 3 shows that finer states always help in expectation, but at the cost of increased estimation variance (Corollary 2). SCX's adaptive state discovery balances this bias-variance trade-off.

5. **Unified framework.** The Jensen perspective (Section 8) unifies the SCX weighting formula with information-theoretic principles: the gap Δ equals the expected reduction in risk from conditioning on the state variable, analogous to the reduction in variance in Rao-Blackwellization.

References

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Changelog

Date	Defect	Change	Severity
2026-06-28	B4	**Added entropy inequality** (§8.1): $H(w_{\text{global}}) \geq \mathbb{E}_s[H(w^*(s))]$ (correct orientation: global entropy is **at least** expected per-state entropy). The original claim $H(w_{\text{global}}) \leq \mathbb{E}_s[H(w^*(s))]$ was reversed — by concavity of entropy, the inequality runs $\geq$, not $\leq$.	MAJOR

Date	Defect	Change	Severity
2026-06-28	B5	**Corrected correlation claim** (§8.2): Replaced the unqualified $\mathbb{E}[(\ell - \hat{R})(w - w')] \geq 0$ with a working conjecture that requires $\mathrm{Cov}(\ell - \hat{R}, w - w') = 0$. Without this additional assumption, the sign is not guaranteed.	MAJOR